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1  Programming
2  -----
3  Program -- set of instructions
4
5  a = 2 -- instruction 1
6  b = 3 -- instruction 2
7  c = a + b; -- instruction 3
8
9  instruction 1 + instruction 2 + instruction 3 = program p1
10
11 a = 2 -- instruction 1
12 b = 3 -- instruction 2
13 c = a - b; -- instruction 3
14
15 instruction 1 + instruction 2 + instruction 3 = program p2
16
17 a = 2 -- instruction 1
18 b = 3 -- instruction 2
19 c = a * b; -- instruction 3
20
21 instruction 1 + instruction 2 + instruction 3 = program p3
22
23
24 p1 + p2 + p3 = software
25 collection of programs is known as software...
26
27 Zoom -- video , audio , screen share , background , chat , waiting room , recording
28 =====>
29 Softwares
30 -----
31 -> collection of programs is known as software...
32 *Types of Software*
33
34 1. System Software
35 --> it works on system and for the system
36 --> Examples : OS (Operating System) --> Windows , MacOS , Linux
37 --> Examples : Languages --> 1. Structured language -- Hindi , Gujarati , english
38                               2. High Level Language -- C , C++ , JAVA , PYTHON
39                               3. Low/ binary/ machine level language (0 and 1)
40
41
42
43 ASCII - American Standard code for information interchange
44 A - 65          a - 97
45 B - 66          b - 98
46 C - 67          c - 99
47
48
49 Task 1 : Convert name characters into ASCII
50 H(72) , a(97) , r(114) , s(115) , h(104)
51 C(67) , h(104) , a(97) , n(110) , d(100) , r(114) , e(101) , s(115) , h(104)
52 K(74) , a(97) , s(115) , h(104) , i(105) , s(115) , h(104)
53
54 Name Characters in ASCII
55 -----

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56 7297114115104 - Harsh
57 6710497110100114101115104 - Chandresh
58 7497115104105115104 - kashish
59
60 Converting ASCII Into Binary
61 Decimal to binary
62
63      A - 65                      A - 01000001
64
65      2 | 65
66      2 | 32 - 1
67      2 | 16 - 0
68      2 | 8  - 0
69      2 | 4  - 0
70      2 | 2  - 0
71      2 | 1  - 0
72
73      C(67) - 01000011           K(74) - 01001011
74      h(104) - 01101000         a(97) - 01100001
75      a(97) - 01100001           s(115) - 01110011
76
77
78 TEST****
79 1. Decode the Message
80     01001000    01101001
81     The message is 'Hi'
82
83 2. Decode the message
84     A superhero's secret initials are:
85     01010011    01010000
86
87 --> Examples : Translator --> converts HLL to BL
88                      --> 1. Compiler
89                      --> checks all code at once and returns all errors at
90                          one time
91                      2. Interpreter
92                      --> checks line by line and returns error line by line
93                      3. Assembler
94                          . Lodaer
95                          . Linker
96
97 2. Application Software
98 --> Softwares which are used for some specific purpose
99 --> Zoom - meetings and calls
100 --> BookMyShow - Book online Tickets of events
101 --> IRCTC - book and view railway details online
102 --> WORD - lets you make a document
103 --> Turbo c - Let's you make c and c++ programs
104
105 Web Application    -- websites (amazon.com , flipkart.com)
106 Mobile Application -- Youtube , Google Drive , Music , Amazon
107 AR (Augmented Reality) , VR(Virtual Reality) , MR(Mixed reality)(HoloLens)
108
109 3. Utility Software
110 --> CAD (Computer Aided Design)
111 --> CAM (Computer Aided Manufacturing)

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110
111 =====>
112 *Extra Information*
113 -----
114
115 0 and 1 - bit
116 8 bit - 1 byte
117
118      128  64 32 16 8 4 2 1
119      0  0  0  0  0  0  0  0    0
120      0  1  0  0  1  0  0  0    64 + 8 = 72 (H)
121      0  1  1  0  1  0  0  1    64 + 32 + 8 + 1 = 105 (i)
122  A    0  1  0  0  0  0  0  1    64 + 1 = 65
123      0  1  0  0  1  0  0  0    64 + 8 = 72 (H)
124      0  1  1  0  0  0  0  1    64 + 32 + 1 = 97(a)
125      0  1  1  1  0  0  1  0    64 + 32 + 16 + 2 = 114(r)
126      0  1  1  1  0  0  1  1    64 + 32 + 16 + 2 + 1 = 115(s)
127
128
129 =====>
130 Homework
131 -----
132 HW 1 : List the languages which uses compiler, interpreter , assembler
133
134 HW 2 : RAM --> random access memory
135        ROM --> read only memory
136        STORAGE --> device storage
137
138        what it is?
139        what does it do?
140        what is the size of them?
141
142 HW 3 : Decode this message
143      01001000    01100001    01110010    01110011
144      H          a          r          s
145
146
147 hw 4 : Convert your full name into binary
148 =====>
149 22/11/2025
150 -----
151
152 Flowcharts and Algorithms
153 -----
154
155 Flowcharts --> diagram used to represent an algorithm
156      Shapes --> specific work
157      start /stop -> oval
158      process -> rectangle
159      Input / output -> Parallelogram
160      Decision -> Diamond
161      Flow -> Arrow
162
163 Algorithms -- step by step instructions used to make flowcharts
164      -- step by step solution of a process/question

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165          -- non - techical , technical
166  =====>
167  Task : Make maggi (non technical)
168
169  step 1 : start
170  step 2 : requires veggies , pan , water , packed maggi
171  step 3 : collect all the ingredients
172  step 4 : boil the water
173  step 5 : add all the ingredients
174  step 6 : stir well
175  step 7 : check if maggi is cooked or not
176          if cooked then go to next step (condition)
177          else go to step 6 (condition)
178  step 8 : serve hot maggi
179  step 9 : end
180
181  Task : Join Harsh Sir meeting....on ZOOM
182
183
184  =====>
185  Technical Algorithms
186
187  1. Make a algo to print sum of two integers
188
189  step 1 : start
190  step 2 : declare variable a, b, c
191  step 3 : input a and b from user
192  step 4 : c = a + b;
193  step 5 : print c (output)
194  step 6 : end
195
196  2. Check entered number is even or not
197
198  step 1 : start
199  step 2 : declare and input num
200  step 3 : if num % 2 == 0 print even
201          else print not even
202  step 4 : end
203
204  3. check entered number is negative or positive
205
206  step 1 : start
207  step 2 : decalre and input num
208  step 3 : if (num > 0) print "positive"
209          else print "negative"
210  step 4 : end
211
212  condition --> true --> output1
213          --> false --> output2
214
215  =====>
216  C Language
217  -----
218  It is known as mother of all lanaguges
219  Developed By : Dennis Ritchie (1972) (53)

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220
221 He is known as father of C Language
222
223 Procedure oriented programming language(function)
224 =====>
225 Structure of C Program
226 -----
227 1. Header Section
228 2. Link Section
229 3. Global Declaration Section
230 4. Main function Section
231 5. User defined function section
232
233 =====>
234 Comments
235 -----
236 Which are ignored by compiler and are used for user information
237
238 1. Single line comment
239 // single line comment
240
241 2. Multi Line Comments
242 /*
243     Author : harsh patel
244     doc : today
245     objetive : first c program
246 */
247
248 =====>
249 1. Header Section
250 -----
251 It contains header files
252 Header Files -> library / inbuilt functions
253
254 # -> preprocessor (it tells the compiler to check if specific file is available or not)
255 include -> directive (directory / folder)
256 <> -> angular brackets
257
258 #include<header_file_name.h>
259
260 #include<stdio.h> // standard input output header file (printf , scanf)
261 #include<conio.h> // console input output header file (clrscr, getch)
262
263 =====>
264 4. Main Function Section
265 -----
266 It is starting and ending point of a program
267 execution starts with and ends with main function only
268
269 return_type function_name(args...)
270 {
271     //body/logic of the program
272 }
273
274 void main()

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```
275  { //body of the function
276      clrscr(); // function calling
277
278      getch(); // function calling
279  } //end of main
280  =====>
281  Task 1 : First C Program
282  -----
283
284  #include<stdio.h>
285  #include<conio.h>
286
287  void main()
288  {
289      clrscr(); // clear the screen
290
291      printf("Hello World!"); // stdio.h
292
293      getch(); // hold the screen
294  } //end of main
295
296  =====>
297  Task 2 : Print hello world and namaste duniya on the screen
298
299  #include<stdio.h>
300  #include<conio.h>
301
302  void main()
303  {
304      clrscr(); // clear the screen
305      // Escape Sequence -- always inside double quotes
306          // \n - new line character
307          // \t - tab line character
308
309      printf("Hello World!\n"); // stdio.h
310      printf("Namsate Duniya!!!");
311
312      getch(); // hold the screen
313  } //end of main
314  =====>
315  HomeWork Section
316  -----
317  1. Make algorithm for subtracting two numbers
318  2. Make algorithm for printing odd or even
319  3. Make algorithm for printing sum of 20 variables
320
321  4. Print your name, age and class on the screen using code
322  5. Print multiplication table of 5 using code
```